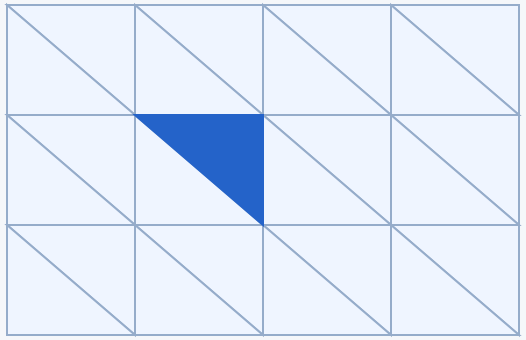
Finite elements → distributed algebra

A brief reminder: local basis functions turn a PDE into a sparse system.

01 Mesh the domain



Small elements, local support

02 Approximate the field

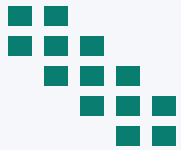
$$u_h = \sum_i u_i \varphi_i$$

DoFs u_i weight basis functions φ_i.

For scalar P1: one DoF per vertex

03 Assemble the system

$$Au = b$$



Sparse coupling between DoFs

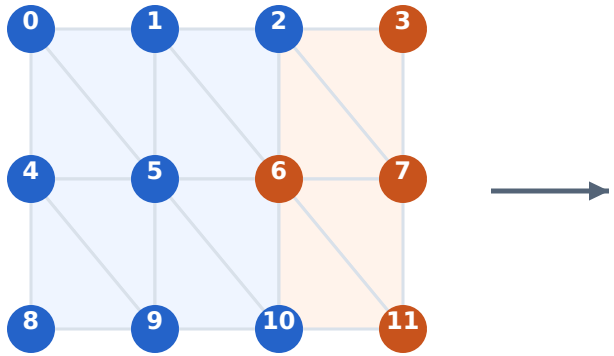
FEniCSx: the platform • DOLFINx: meshes, function spaces and distributed assembly

Local finite-element support creates sparse algebra with distributed ownership.

One mesh, two MPI ranks

Scalar P1 • colors identify the owner • outlines identify local ghost copies

GLOBAL CELL PARTITION



Illustrative ownership;
actual IDs come from IndexMap.

Rank 0 Rank 1

RANK 0 / 8 owned + 1 ghost

[owned values | ghost copies]

RANK 1 / 4 owned + 2 ghosts

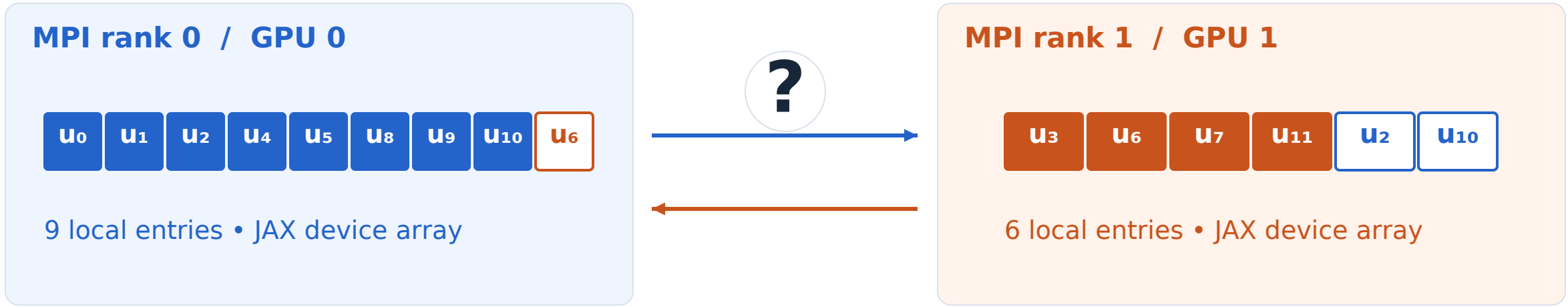
[owned values | ghost copies]

Each DoF has one owner. Other ranks keep copies where local work needs them.

Ghost DoFs are copied coefficients; ghost cells are additional mesh cells (not shown here).

How do we preserve this on GPUs with JAX?

Keep the FEM ownership model. Move repeated numerical work onto the devices.



Preserve ownership

Reuse DOLFINx's owner/ghost map.

Communicate inside JIT

Refresh ghosts between GPU steps.

Handle unequal sizes

Partitions need different storage.

How do remote owner values reach the right local ghost slots?

A local matrix row needs a remote vector value

A four-DoF algebraic toy • row partitioning • zero-based indices • start with y = 0

1 SPLIT THE MATRIX BY OWNED ROWS

	A				x	Ax
0	2	−1	0	0	1	0
1	−1	2	**−1**	0	2	−1
2	0	**−1**	2	−1	4	−2
3	0	0	−1	2	8	12

The matrix stays local; selected x values travel.

2 EXCHANGE x, THEN MULTIPLY LOCAL ROWS

Rank 0 needs remote x_2

1 | 2 | 4

$$y_1 = -1\cdot 1 + 2\cdot 2 - 1\cdot 4 = -1$$

Rank 1 needs remote x_1

4 | 8 | 2

$$y_2 = -1\cdot 2 + 2\cdot 4 - 1\cdot 8 = -2$$

A remote column reference becomes a local ghost-vector lookup.

Our API computes owned y += Ax; x and matrix inputs are preserved, and output ghosts stay unchanged.

mpi4jax: pack → sendrecv → unpack

The ownership plan is prepared once; the JIT-compiled operation exchanges current values.

1 Pack owner entries

1 2 ? send 2

`x[send_indices]`

2 Exchange with peers

`mpi4jax.sendrecv`

Compiled bridge → MPI

Neighbor buffers, fixed sizes

3 Fill ghost slots

1 2 4

ghost slots ← received

GPU buffers can follow different transport paths

HOST-STAGED

GPU → CPU → MPI → CPU → GPU

CUDA-AWARE MPI

GPU pointer → MPI → GPU pointer

CUDA-aware MPI accepts device buffers; the actual transport may still stage on the host.

The challenge: make the whole HPC stack agree

Python packages, native libraries and the selected runtime transport must be compatible.

01 CUDA discovery across Spack + venv

mpi4py and CUDA lived in different prefixes.
The mpi4jax build needed a discovery fix.

02 Selected UCX lacked GPU support

The loaded transport misread a device buffer.
A working MPI import did not validate this path.

Successful installation is only the start: validate the GPU exchange in the actual job.

Explicit ghost exchange with JAX sharding

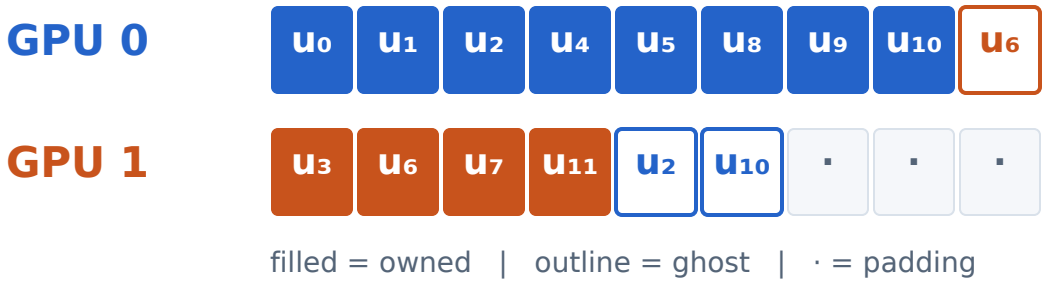
One rank per GPU • retain DOLFINx ownership • let JAX execute the numerical collective

ONCE / HOST

DOLFINx IndexMap + mpi4py metadata → packing indices, receive slots, masks

Initialize distributed JAX; create a one-dimensional device mesh with axis “rank”.

GLOBAL ARRAY SHAPE [2, 9] / ONE ROW PER GPU



Explicit packets preserve the ghost map

GPU 0 sends [u_2, u_{10}] → GPU 1 ghost slots

GPU 1 sends [u_6, 0*] → GPU 0 ghost slot

* Fixed-width packet padding is masked / discarded.

EACH JIT-COMPILED CALL / shard_map runs the same body on every device

Pack by indices → `lax.all_to_all` → **Scatter to ghosts** → **Local CSR matvec**

Sharding places the arrays. Our plan defines which owner values fill which ghosts.

Current scope: forward INSERT + scalar CSR matvec. Tradeoff: padding and a global all-to-all collective.